

ENTERPRISE INTERNSHIP REPORT

Building a Text Scanning Platform Integrated with OCR Technology at HiCAS Company

DocuGraph RAG Platform - Application of Gemini AI Model & Graph Database

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OVERVIEW OF HICAS SOFTWARE COMPANY

Professional IT Outsourcing Department

- **Operational Model:** Accompanying from on-site survey to optimal software solution design, comprehensive management, and operations.
- **Operational Goal:** Become a trusted technology partner, focusing on practical solutions based on real-world experience.
- **Core Strengths:** Plant diagram design (P&ID), BIM/CAD, hospitality & tourism, AI data analytics, and logistics optimization.
- **Human Resources:** Experienced engineering team with high technical expertise, meeting strict domestic and international demands.

ViTHEP

Specialized 3D steel structure information modeling solution for power transmission poles and industrial plants; seamlessly integrated with Revit and BIM

SmartMTO

AI-powered product suite analyzing project data, automatically extracting 2D/3D drawings from scans, PDFs, and CAD files (DWG, IFC)

AnyOn

Common Data Environment (CDE) solution for centralized construction management, enabling effective BIM project collaboration.

TingConnect

BIM application solution for smart building operations, integrating resident app, operational management & services.

CURRENT STATE AND EXISTING CHALLENGES

Distributed Knowledge Management Challenge

- **Diverse Formats:** Information is scattered across various sources including PDF documents, scanned images, and web articles.
- **High Processing Time:** Users waste excessive effort reading, manually searching keywords, and synthesizing information from massive document sources.
- **Locked Information:** Much valuable internal knowledge is trapped in image files and scanned PDFs that traditional keyword search engines cannot access

Critical Limitations of Public AI

- **Hallucination Risk:** Models generate responses based on probability, easily fabricating unrealistic information when lacking accurate data.
- **Lack of Internal Knowledge:** Public AI has no access to private documents, specialized knowledge, or internal corporate reports.
- **Sensitive Data Leakage:** Uploading confidential documents to third-party servers poses massive information security risks.
- **Lack of Verification Mechanism:** Does not specify which document or page number the answer was extracted from, preventing user verification

PROJECT OBJECTIVES AND SCOPE

Platform Development Objectives

- **Unified Knowledge Extraction:** Build a workspace (Notebook Workspace) to centrally manage and standardize multi-source documents.
- **Hallucination-Resistant AI Q&A:** Apply RAG (Retrieval-Augmented Generation) combined with Google Gemini LLM to deliver absolutely accurate responses.
- **Knowledge Graph Visualization:** Automatically analyze and represent semantic relationships between entities as interactive, dynamic graphs.

Research Scope & Challenges

- **Input Data Processing:** Focus on handling standard PDFs, captured/scanned images (PNG/JPG), and dynamic web links.
- **Language Support:** OCR text recognition, semantic analysis, and conversational interaction optimized for both Vietnamese and English.
- **Specialized Focus:** Optimize context chunk retrieval accuracy (Cosine Similarity) and display direct citations at the page level.

TECHNOLOGY STACK USED

FRONTEND & UI CORE

- **Next.js 15 & React 19:** Modern web development framework, UI optimization, and server-side rendering.
- **Tailwind CSS & Tailwind Elements:** Build responsive, minimalist, and smooth user interfaces.
- **Cytoscape.js / Canvas:** Real-time interactive knowledge network graph visualization.

BACKEND FRAMEWORK

- **FastAPI ASGI:** High-speed asynchronous web framework delivering high-performance RESTful APIs.
- **Python 3.12+:** Core language for logic processing, scientific computing, and data flow orchestration.
- **LangChain Framework:** RAG orchestration framework connecting external data and managing LLM prompt chains.

AI MODEL & PROCESSING

- **Google Gemini 2.5 Flash:** Smart processing large language model with ultra-fast streaming responses.
- **text-embedding-004:** Model for generating high-quality 768-dimensional text vector embeddings.
- **PaddleOCR Deep Learning:** High-efficiency OCR toolkit for Vietnamese text recognition.

DATABASES & INFRASTRUCTURE

- **PostgreSQL & pgvector:** Relational database storing metadata combined with vector extensions for semantic search.
- **Neo4j Graph Database:** Graph DB managing links, storing entities, and knowledge relationships.
- **Docker & Docker Compose:** Containerization packaging for easy configuration, maintenance, and deployment

MULTI-SOURCE KNOWLEDGE EXTRACTION PIPELINE

STEP 1: INGESTION

- **Multi-page PDF:** Upload standard PDF files with automatic parsing and page count analysis.
- **Scanned/Captured Images:** Recognize and extract text from PNG/JPG image files.
- **Web URL:** Input static/dynamic web links with automated main content scraping.



STEP 2: EXTRACTION & CHUNKING

- **PyMuPDF:** Extract high-quality raw text from PDF documents.
- **PaddleOCR:** Vietnamese/English character recognition from images using Deep Learning models.
- **Recursive Splitter:** Split text into smaller chunks with appropriate overlapping context.



BƯỚC 3: VECTOR & GRAPH

- **Gemini Embedding:** Embed text chunks into 768-dimensional vector arrays.
- **pgvector:** Store vector arrays and chunk metadata into PostgreSQL.
- **Neo4j Graph:** Extract entities and relationships, storing them as network diagrams.

DUAL DATABASE ARCHITECTURE

Vector DB (PostgreSQL + pgvector)

- **Relational Metadata Management:** Store user account info, Notebook structures, and detailed chat histories.
- **Similarity Vector Storage:** Store 768-dimensional embedding vectors representing chunk semantics.
- **Cosine Distance Queries:** Perform mathematical distance comparisons to retrieve chunks with the highest relevance.
- **Data Integrity:** Ensure Row-Level Security (RLS) rules, relational normalization, and foreign key constraints

Graph DB (Neo4j Graph Database)

- **Physical Document Modeling:** Store Notebook, Source, and Chunk nodes to accurately reflect source document structures.
- **Knowledge Network Extraction:** Manage entity nodes (Person, Organization, Location, Concept) and semantic relationships.
- **GraphRAG Context Queries:** Combine semantic vector search with adjacent entity relationships on the knowledge graph.
- **Superior Performance:** Optimize deep relationship queries (n-hops) using specialized Cypher Query language.

PROPOSED BREAKTHROUGH TECHNICAL SOLUTIONS

MULTI-TIER HYBRID SEARCH:

- **Semantic & Keyword Search:** Combine Vector Similarity with BM25 Full-text indexing.
- **RRF (Reciprocal Rank Fusion) Algorithm:** Recalculate ranking scores to maximize balance between semantic understanding and exact keyword precision.

GRAPH ENTITY RESOLUTION:

- **Neo4j Node Optimization:** Prevent graph fragmentation caused by synonyms or abbreviations creating isolated nodes.
- **Automated Entity Merging:** Compare vector distance (score ≥ 0.85) combined with Gemini API to normalize entities and generate alias attributes

PAGE-LEVEL DETAILED CITATIONS:

- **Accurate Metadata Recording:** Store page numbers and bounding box coordinates for each text chunk.
- **Interactive Direct PDF Viewer:** Clicking a citation card automatically opens the original PDF page and highlights the contextual text inside Next.js.

AUDIO PODCAST AI GENERATION

- **Creative Knowledge Transformation:** Convert entire Notebook content into a 2-host Vietnamese conversational audio podcast.
- **Script Generation Flow:** Use Gemini 2.5 Flash to generate natural dialogue combined with multi-voice Text-To-Speech.

SYSTEM USE CASE DIAGRAM

Role: System Administrator (Admin)

- **User Account Management:** View list, search, control access permissions, lock or unlock user accounts in the system.
- **Hardware Resource Monitoring:** Track Vector DB storage capacity, number of initialized Notebooks, and Google Gemini API calls.
- **System Parameter Configuration:** Manage Google Gemini API keys, RAG parameters (Chunk size, Chunk overlap), and database connections.

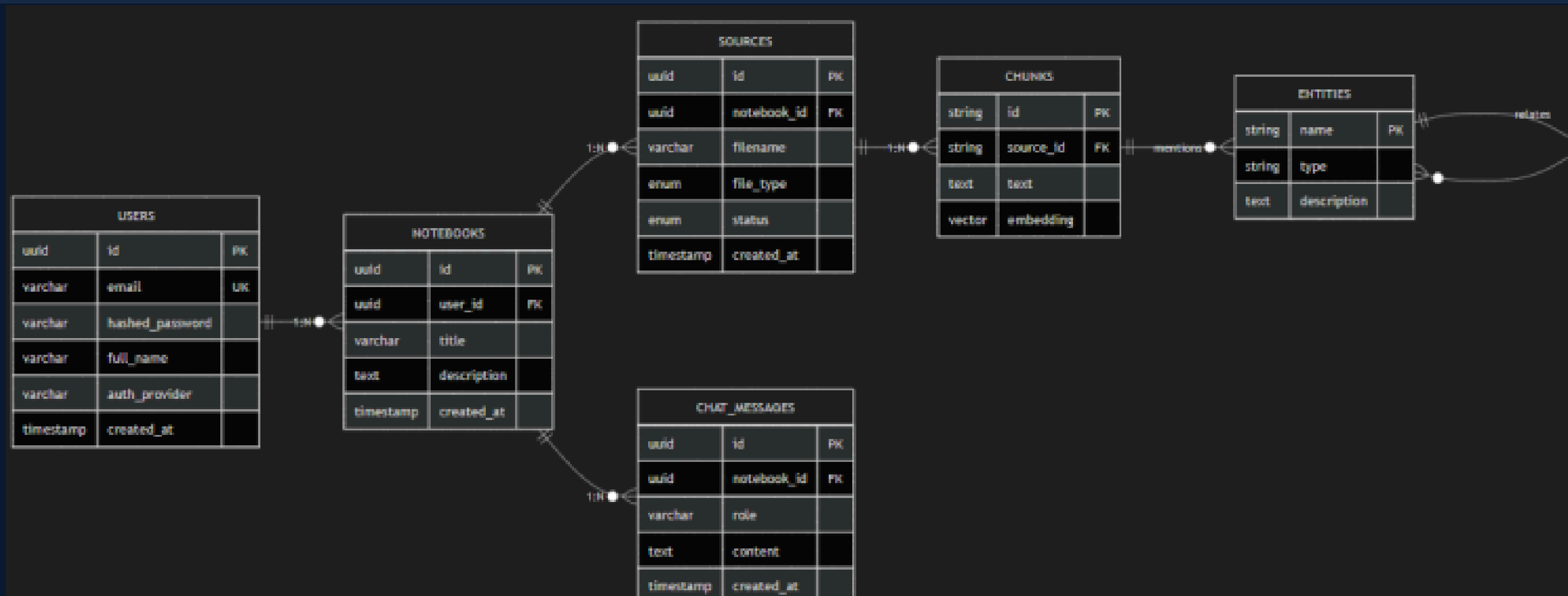
Role: End User (User)

- **Notebook Space Management:** Create specialized notebooks, edit titles and descriptions, or delete workspaces.
- **Multi-Source Knowledge Management:** Upload multi-page PDFs, OCR captured images, or enter web links for text extraction.
- **Interactive AI Chat Q&A:** Select data sources, ask questions, receive streaming responses, and verify via citation cards.
- **Explore Knowledge Graph:** Interact with entity networks, zoom in/out, drag-and-drop, and trace back to original source text chunks.

DUAL DATABASE MODEL (ERD)

PostgreSQL (Relational Database)

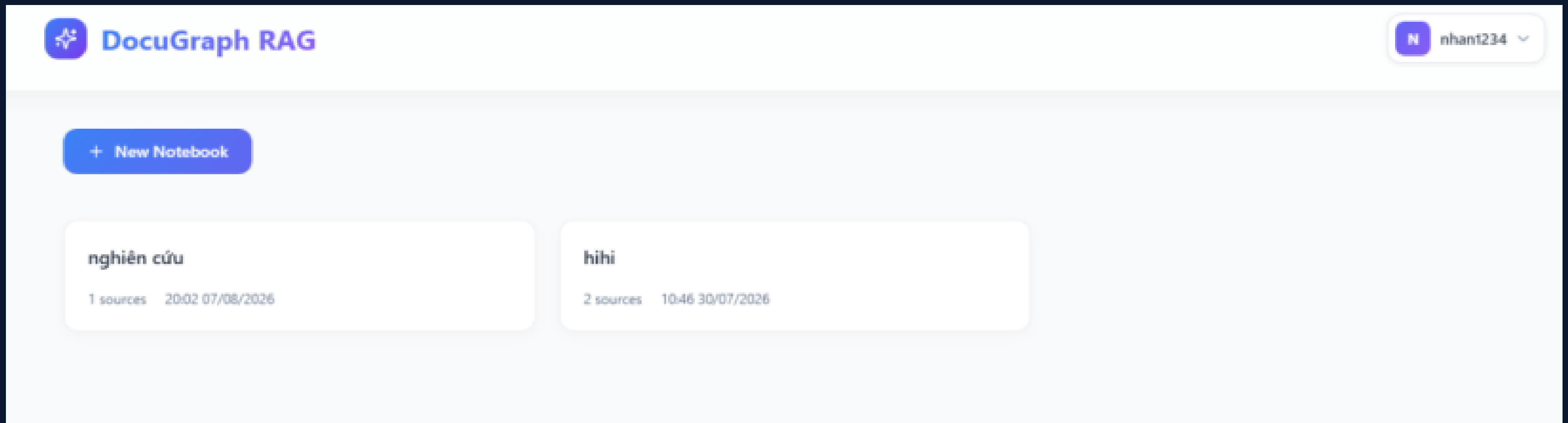
- **Users Table:** Stores id (UUID), login email, Bcrypt-hashed hashed_password, full_name, avatar_url, and authentication method (local/google).
- **notebooks Table:** Manages personal notebook spaces, user_id foreign key referencing users, title, description, and creation timestamp.
- **sources Table:** Stores id, notebook_id (FK), filename, file_type, status ('processing', 'ready', 'error'), page_count, preview, and web url.
- **chat_messages Table:** Stores chat history, notebook_id (FK), role ('user', 'assistant'), content, and sources_used as a JSON array storing references.



DEMO: WORKSPACE UI & DATA INGESTION

Smart Workspace

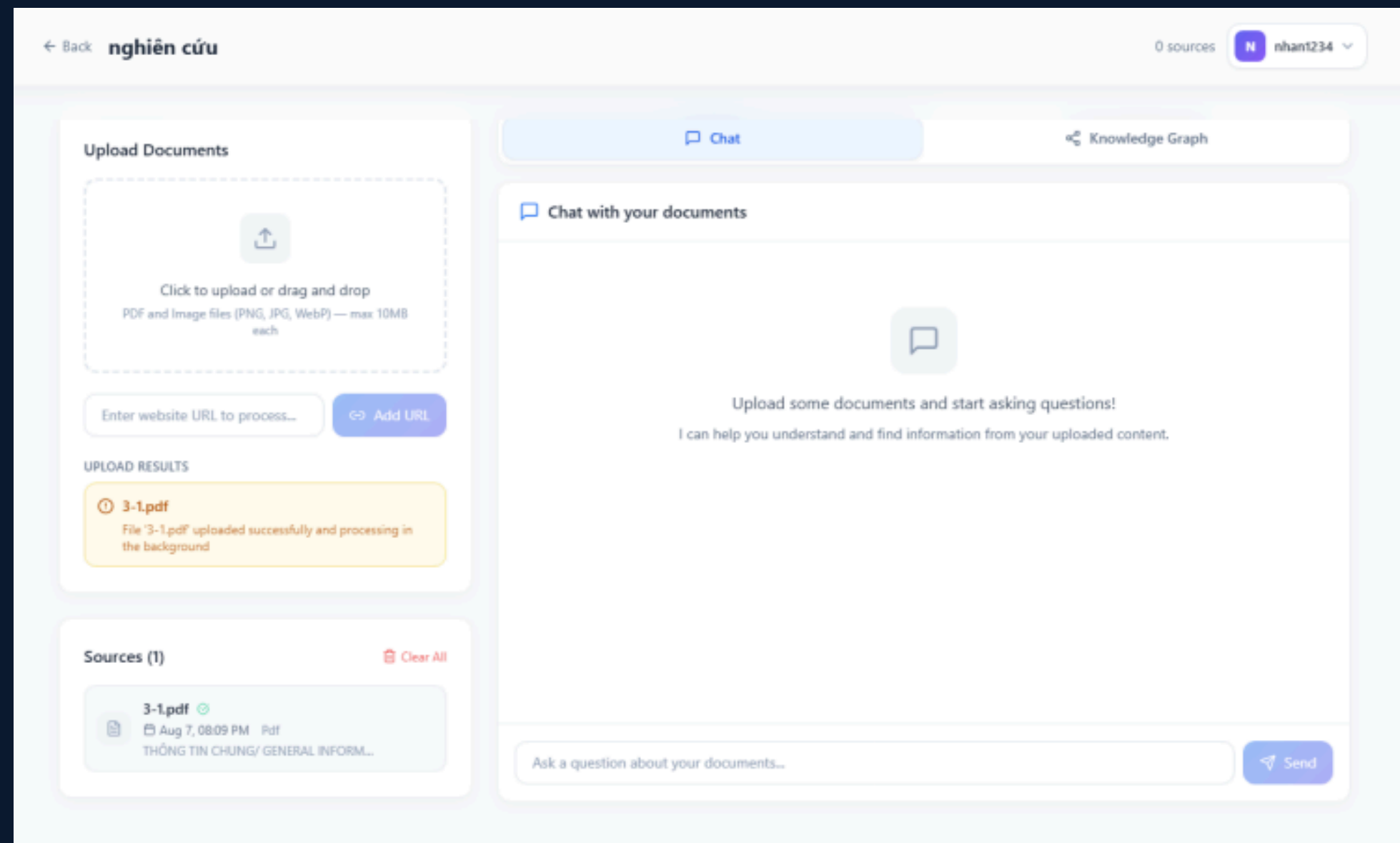
- **Modern Glassmorphism Design:** Smooth interface over soft gradient backgrounds, smooth hover effects, and vivid visualizations.
- **Notebook Management Dashboard:** Centralized screen displaying notebooks as cards, making project tracking and grouping easy.
- **Scientific 2-Column Layout: Clear separation:** Left column handles document upload and source management; Right column provides flexible tab switching between AI Chat and Knowledge Graph.



DEMO: WORKSPACE UI & DATA INGESTION

Automated Data Ingestion Pipeline

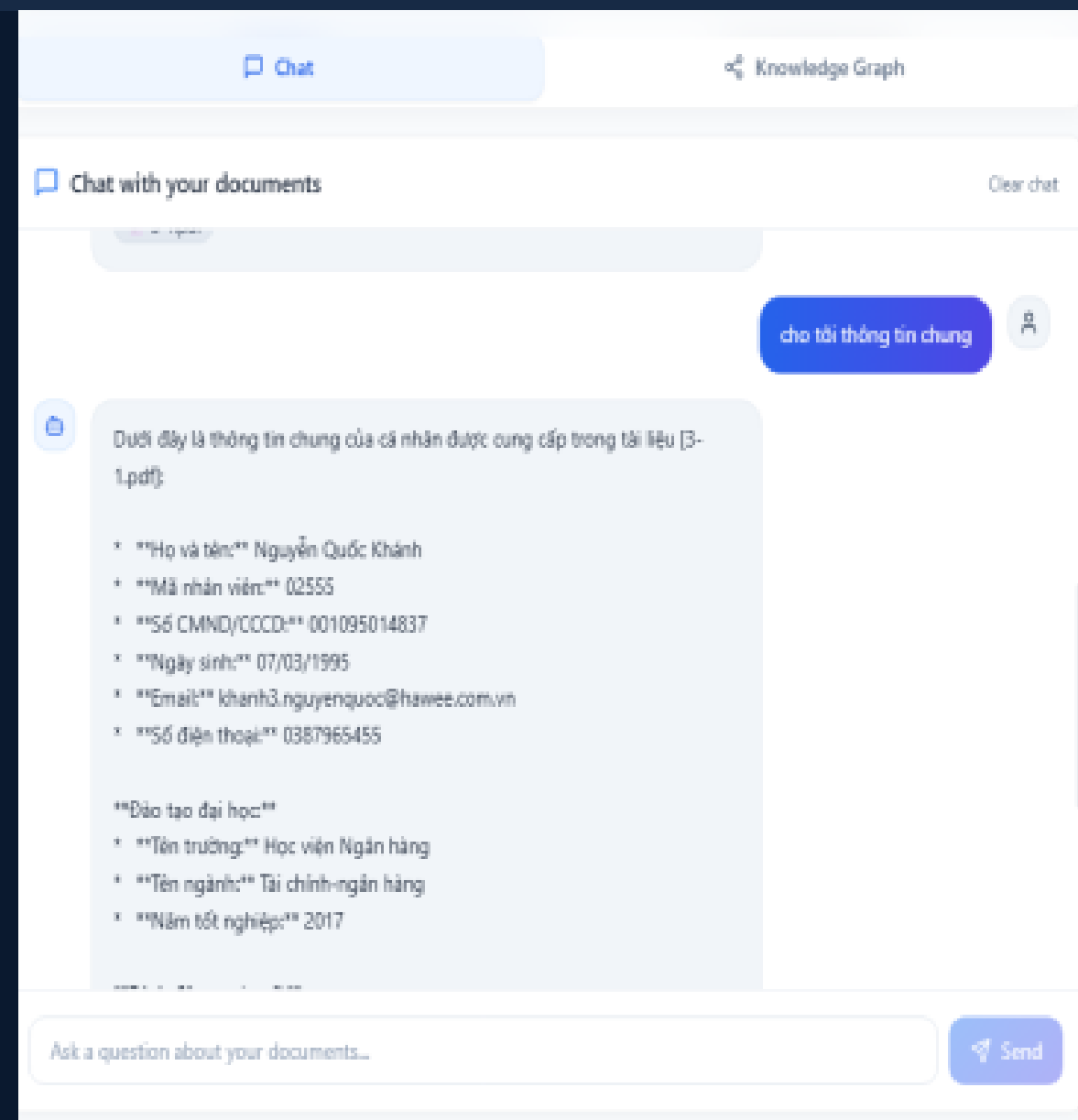
- **Drag-and-Drop PDF Support:** Automatically parses raw page structures at ultra-high speed using PyMuPDF.
- **In-Depth Image OCR:** Integrated PaddleOCR deep learning model for high-accuracy Vietnamese handwriting/photo recognition.
- **Automated Web Article Scraping:** Strips redundant HTML tags and advertisements to extract clean text for RAG.
-



EMO: REAL-TIME AI Q&A

TAI RAG Conversational Interaction

- **Strict Document-Grounded Q&A:** AI responses are strictly synthesized based on active documents within the current Notebook.
- **Streaming SSE Mechanism:** FastAPI streams Server-Sent Events data, displaying responses word-by-word in real time.
- **Gemini 2.5 Flash Model:** Smart assistant providing deep analysis, understanding complex query intents, and responding extremely fast



DEMO: REAL-TIME AI Q&A

Citation & Source Verification Mechanism

- **Detailed Citation Cards Below Answers:** Clearly attaches source file name and precise page number (e.g., [Source: 3-1.pdf, Page 2]).
- **Interactive Frontend Document Viewer:** Clicking citation tags immediately opens a viewing window displaying the PDF / extracted raw text side-by-side.
- **Auto-Scroll & Context Highlighting:** Automatically scrolls to the document page containing the info and highlights the exact original paragraph used by AI.

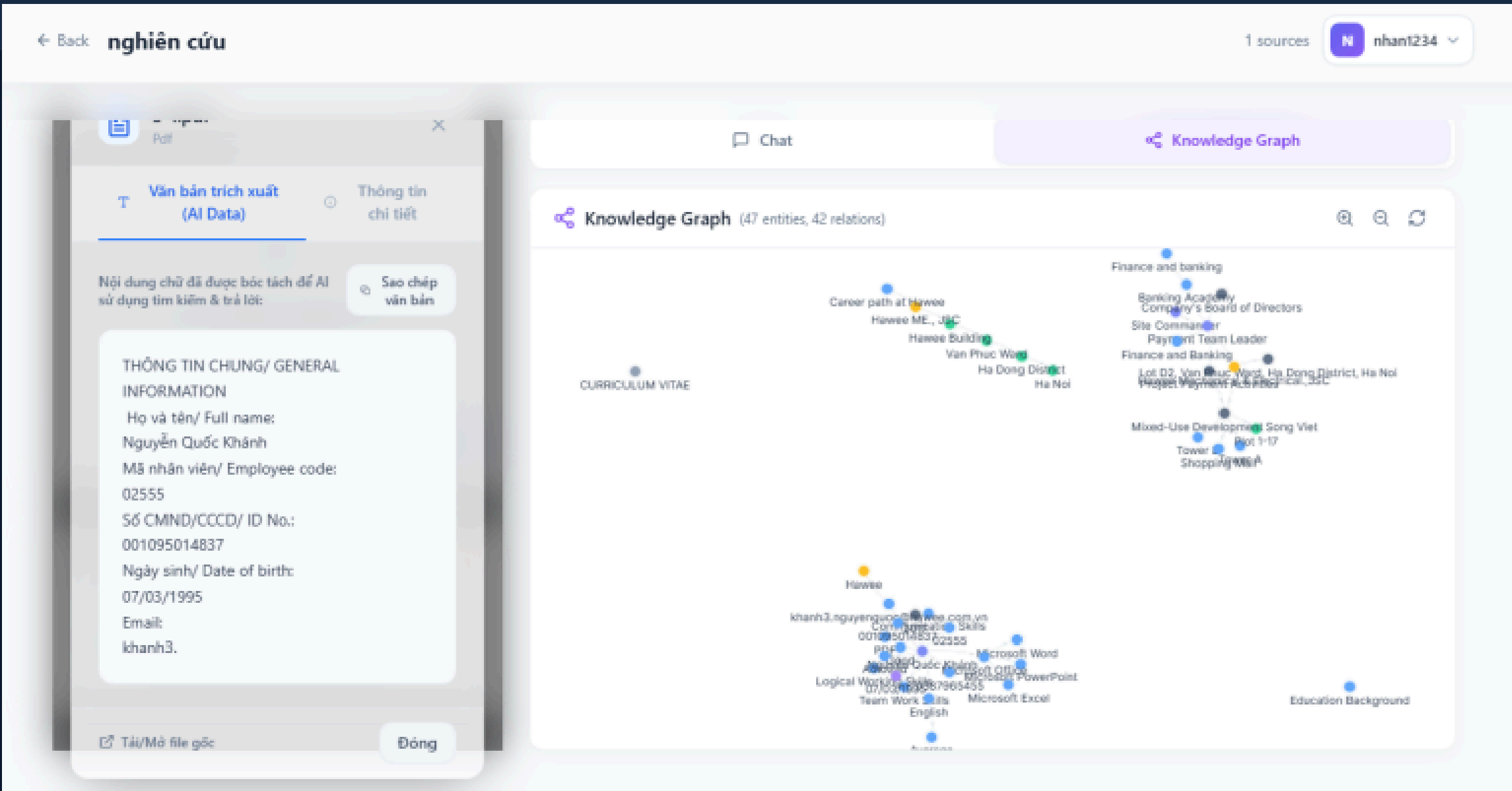
Họ và tên / Full name:		Nguyễn Quốc Khánh		Mã nhân viên / Employee code:	00555
Số CMND/CCCD ID No.:		0010000100037		Mẹ sinh / Date of birth:	07/03/1988
Email:		khanh.q.nguyen@hcmu.edu.vn		Số điện thoại / Phone number:	0907665455
Đào tạo và lý năng / (Mẫu có chứng chỉ hành nghề thì chỉ liệt kê thông tin chứng chỉ hành nghề, không liệt kê thông tin chứng chỉ bồi dưỡng nghiệp vụ) Education Background:					
<u>Đào tạo đại học / University Education:</u>					
- Tên trường / University:	Học viện Ngân hàng / Banking Academy				
- Tên ngành / Major:	Tài chính ngân hàng / Finance and banking				
- Năm tốt nghiệp / Year of Graduation:	2017				
<u>Trình độ ngoại ngữ / Language skills:</u>					
- Tên chứng chỉ / Certificate name:					
- Tên ngoại ngữ / Language:	Tiếng Anh / English				
- Nghĩa, số, đọc, viết / Level:	Trung Bình / Average				
<u>Kỹ năng vi tính / Computer skills:</u>					
- Thành thạo các phần mềm / Proficient in using software:	Autocad, Microsoft Office (Word, Excel, Power Point), PDF				
<u>Kỹ năng giao tiếp / Communication skills:</u>					
Tốt / Good					
<u>Kỹ năng làm việc logic / Logical working skills:</u>					
Tốt / Good					
<u>Kỹ năng làm việc theo nhóm / Team work skills:</u>					
Tốt / Good					
Kinh nghiệm làm việc (sắp xếp theo thứ tự thời gian làm việc gần đây nhất) / Work Experience					
STT No.	Thời gian Period	Công ty Company	Địa chỉ Address	Nội dung hoạt động Details	
Quá trình làm việc tại Hạng / Career path at Hạng					
1	05/2022 - nay	Công ty Cổ phần Hạng Cơ Điện Hạng Mechanical & Electrical JSC	Lô D1 Khu Đô thị Đầm Trạc (Phước, Hà Đông, Hải Phòng) Lot D1, Van Phuc ward, Ha Dong district, Hai Noi	Dự án Project: Địa điểm Address: Chức năng dự án Purpose of use: Quy mô dự án Scale of project: Cấp công trình Project level: Chức danh Title:	Nhu cầu hợp tác tiếng Việt, L1-17 Mixed-Use Development Song Viet Plot 1-17 Plot 1-17, Khu đô thị mới Thủ Thiên, Tp. Thủ Đức, Tp. Hồ Chí Minh TTTM Shopping Mall Gồm: 2 tháp A & B, mỗi tháp cao 23 tầng Includet: Tower 2 towers A & B, each tower is 23 floors Cấp I Level I Trưởng nhóm thành viên Represent Team Leader
Công việc:				Thực hiện các nhiệm vụ do Ban giám đốc Công ty, Chủ hay Trưởng Ủy quyền. Thay mặt Chủ hay Trưởng giải quyết và chịu trách nhiệm các công việc được giao trong công tác thành viên của dự án	

The screenshot displays the Microsoft Copilot chat interface. At the top, there's a header bar with a 'Chat' button on the left and a 'Knowledge Graph' icon on the right. Below the header, the main chat area shows a conversation. The chat history on the left includes a message from the user asking for a summary of a document. The current chat message, sent by the user, asks for personal information from a document. The response from Copilot is a structured list of personal details extracted from a PDF file, including name, ID, birth date, email, phone number, education, and work experience. The interface is clean and modern, with a light blue and white color scheme.

DEMO: KNOWLEDGE GRAPH VISUALIZATION

Knowledge Network Map

- **Network Graph Visualization:** Uses Cytoscape.js library to render entity nodes and relationship links connected with Neo4j.
- **Entity Distinction by Color:** Color-coded UI scheme (e.g., Person - Green, Organization - Blue, Location - Orange...).
- **Visual Semantic Relationships:** Clearly represents link types between nodes (EMPLOYED_BY, LOCATED_IN, PART_OF...).



CONCLUSION AND FUTURE DEVELOPMENT

Achievements & Limitations

- **Product Success:** Fully built DocuGraph RAG platform enabling multi-source knowledge extraction, smooth streaming AI Q&A with citations.
- **Effective Graph Visualization:** Completed entity extraction, Neo4j storage, and dynamic interactive network rendering on the Frontend.
- **Practical Skill Acquisition:** Adapted well to HiCAS corporate environment, mastered AI RAG, Vector/Graph DB, and OCR technologies.
- **Existing Limitations:** Large file OCR pipeline consumes high server resources; Neo4j entity deduplication algorithm remains basic.

Future Development Orientations

- **Asynchronous Queue Integration:** Deploy Celery architecture with Redis broker to offload heavy tasks (OCR, Embedding, Neo4j) to Workers.
- **Dynamic Web Scraping Upgrade:** Replace BeautifulSoup with Playwright Engine running headless browsers to improve input data quality.
- **Hybrid Search & RRF Synchronization:** Deploy vector and BM25 hybrid search mechanism into actual production environment.
- **HiCAS Practical Application Proposals:** Request company GPU server support to test parsing complex CAD/BIM drawings and large project dossiers.

Thanks!